

Assignment 2

Due Date: February 19, 2025 at 7:30 PM

Overview

This assignment will be worth **4.5%** of your total grade. Parts 1 and 2 are designed to utilize the asynchronous properties of Node.js. Part 3 is a way to try out a new, fun package.

Important: If you are on a Windows machine, do not take into account `/r` for anything, or you will lose 10%. If you use the ES6 import statement **instead** of the CommonJS `require` for node modules taught during lecture, you will lose 10%.

Specifications

For this assignment, you will need to understand the asynchronous properties of Node.js.

Part 1: Asynchronous Newline Count

Write a program called `asyncNewlines.js` that uses a single asynchronous filesystem operation to read a file and print the number of newlines it contains to the console (`stdout`).

You will have a file name as the third argument just like in assignment 1.

Hint: Instead of `fs.readFileSync()` you will want to use `fs.readFile()` and instead of using the return value of this method you need to collect the value from a callback function that you pass in as the second argument.

An example run would be: `node asyncNewlines.js test.txt`

Part 2: File Count

Write a program called `asyncFileCount.js` that prints the number of files with a given extension in a given directory. The first argument will be the path to the directory we want to filter on (e.g. `"/path/to/dir/"`) and a file extension to filter by as the second argument.

For example, if you get `'.txt'` as the second argument then you will need to filter the list to only files that end with `.txt`

Below are some example runs:

An example run would be: `node asyncFileCount.js . .js`

Another example run would be: `node asyncFileCount.js /Users/jalirani/Desktop.docx`

Part 3: LoDash

Lo-Dash is a Javascript library that gives a lot of assistance when working with Arrays, JSON and Javascript objects. Lo-Dash can be used with Node.js in the back-end but also in a browser in the front-end. As a preamble, here are some useful functions to know:

- `_.groupBy(collection, callback)`
- `_.map(collection, callback)`
- `_.uniq(collection)`
- `_.orderBy(collection, [fieldName], [orders])`

For our assignment, we have an array of Reddit comments in a JSON file. Below is an example (note: this is just an example structure of a JSON object `{username: some_username, comment: some_comment}`). It could extend beyond or below the seven objects given.

```
[ { username: "nftGeek", comment: "bayc is awesome!" },
  { username: "nftGeek", comment: "do you own one?" },
  { username: "maceth", comment: "my wallet has 0 eth" },
  { username: "max", comment: "crypto is a ponzi scheme" },
  { username: "max", comment: "i love crypto" },
  { username: "maceth", comment: "my wallet has 0 usdc" },
  { username: "max", comment: "i really love your site" } ]
```

Create a file called `reddit.js` that reads in a JSON file and:

- Returns an array with every username for every comment. Note: include repeats.
- Displays each user name as a key with an array of the JSON(`{username: some_username, comment: some_comment}`) as the value
- Returns an array with every unique username for every comment. Note: do not include repeats.
- Return the original JSON object with usernames in ascending order

** You must use at least one LoDash function for each of the four parts. No answer to each of the four parts can be above **three** lines of code, otherwise points will be subtracted.

Example output given the JSON object above:

```
Part 1:
[
  'nftGeek', 'nftGeek',
  'maceth',  'max',
  'max',    'maceth',
  'max'
]
-----
Part 2:
{
  nftGeek: [
    { username: 'nftGeek', comment: 'bayc is awesome!' },
    { username: 'nftGeek', comment: 'do you own one?' }
  ],
  maceth: [
    { username: 'maceth', comment: 'my wallet has 0 eth' },
    { username: 'maceth', comment: 'my wallet has 0 usdc' }
  ],
  max: [
    { username: 'max', comment: 'crypto is a ponzi scheme' },
    { username: 'max', comment: 'i love crypto' },
    { username: 'max', comment: 'i really love your site' }
  ]
}
-----
Part 3:
[ 'nftGeek', 'maceth', 'max' ]
-----
Part 4:
[
  { username: 'maceth', comment: 'my wallet has 0 eth' },
  { username: 'maceth', comment: 'my wallet has 0 usdc' },
  { username: 'max', comment: 'crypto is a ponzi scheme' },
  { username: 'max', comment: 'i love crypto' },
  { username: 'max', comment: 'i really love your site' },
  { username: 'nftGeek', comment: 'bayc is awesome!' },
  { username: 'nftGeek', comment: 'do you own one?' }
]
```

Submission

Please submit the following on Blackboard (may need submit as a .zip):

- asyncNewlines.js
- asyncFileCount.js
- reddit.js